

LESSON 4: INFORMATION SYSTEMS

In previous lessons we have explored how humans interact with their computers and how these computers communicate with each other across networks. We have explored how functions such as input, output, storage and processing can occur over networks as in the case of the cloud and the IoT. In this lesson, we will explore how all these can be brought together, humans, computers, networks, cloud and IoT, to form larger systems to handle information.

FUNCTIONS OF INFORMATION SYSTEMS

Every information system serves a particular purpose, such as managing a school or providing navigation services. Regardless of the purpose, all information systems collect data, store data, and process data in order to extract meaningful information and share it to help guide decision-making.

**AN INFORMATION SYSTEM PERFORMS FOUR PRIMARY FUNCTIONS:
DATA COLLECTION, STORAGE, PROCESSING, AND SHARING.**

An information system performs four primary functions: data collection, data storage, data processing, and information sharing.

Collection: An information system gathers data from various sources. This could be through direct user input or automated data collection from sensors as in the case of the IoT.

Storage: The collected data is stored mainly in the form of databases on servers making the data accessible over a network. Storage may be on a local server or it may be on a server based in the cloud.

Processing: This is where the data is transformed into useful information. The system might organize the data, perform calculations, apply various algorithms, or create visualizations. Depending on where the data is stored, processing may occur at the level of a local server or in the cloud. The aim is to turn raw data into meaningful information that supports decision-making or other activities.

Sharing: Information systems are designed to distribute and share data among users. This sharing ensures that the necessary data is available when and where it's needed. Information systems have multiple users with whom information is shared. Access to the information may be restricted to ensure the security and confidentiality of data and information.

To better understand information systems let's explore two types: a management information system and a geographic information system.

SCHOOL INFORMATION SYSTEMS

A school information system is a system designed to collect, store, process and share information pertinent to all stakeholders in a school, from students and their parents to employees and their administrators.

EXAMPLE 4.1

We will explore a SIS from the perspectives of a student, parent, teacher, principal and the SIS administrator.

Story 1: Student - Sara

Sara is a student in seventh grade. Every morning, she checks her SIS mobile app to see her schedule for the day. After school, she checks for assignments her teachers have posted. Once she completes her homework, she uses the SIS to submit her assignments. When she finishes a big project or a test, she eagerly awaits the grade on the SIS - it's the first place her grades are posted.

In addition, Sara has a smart ID card. She uses it to swipe in and out of school

that way the SIS automatically takes attendance. She also uses it to purchase food and drinks from the school's cafeteria.

Story 2: Parent - Ahmed

Every morning, Ahmed checks his children's timely arrival to school using his SIS mobile app. At work, he frequently checks their schedules and academic progress using his desktop. One day, he noticed a dip in his son's math scores. He messaged the teacher via SIS, and arranged a virtual meeting to discuss improvement strategies. School announcements are easily accessed on SIS, keeping Ahmed informed. Despite his hectic schedule, SIS enables Ahmed to stay closely connected with his children's education.

Story 3: Teacher - Mr. Johnson

Mr. Johnson starts his day by checking his class rosters on the SIS, noting any changes. Throughout the day, he uses the SIS to take attendance, input grades, and assign homework. He appreciates how he can quickly message students or parents through the SIS to discuss grades or concerns. At the end of the day, Mr. Johnson uses information from the SIS to monitor students' progress and customize his lesson plans accordingly.

Story 4: Principal - Mrs. Gomez

Mrs. Gomez begins her day by reviewing reports from the SIS, looking at attendance records, academic performance, and other key metrics. The SIS helps her identify patterns, such as if a particular class is struggling with attendance or a specific subject. She also uses the SIS to communicate with the school community, whether it's a reminder about an upcoming event or a critical update.

Story 5: System Administrator - Mr. Lee

Mr. Lee oversees the SIS, ensuring it runs smoothly. His morning starts with system checks and addressing any issues. Throughout the day, he manages user accounts, granting necessary access to new staff, or students. Mr. Lee also installs updates and troubleshoot issues. He plays a crucial role in training staff on how to use the SIS effectively.

GEOGRAPHIC INFORMATION SYSTEMS

A Geographic Information System (GIS) is a type of information system that is specifically designed to collect, store, process, and share geographical data. GIS allows users to visualize data primarily in the form of maps. It also provides users with information about various locations on the map, such as names of streets, traffic congestion, photographs, and directions to go from one place to another.

EXAMPLE 4.2

Google Maps is a commonly used geographic information system. We will explore it from the perspectives of a user, data analyst, street view driver, 3D modeling specialist and software engineer.

Story 1: User - Sofia

Sofia uses Google Maps in her everyday life. She starts her day by checking the fastest route to her office, taking into account real-time traffic updates. During lunch, she uses Google Maps to find a new restaurant nearby. After work, she uses Street View to virtually explore a neighborhood where she's considering renting a new apartment. For Sofia, Google Maps and its GIS underpinnings make her daily life more convenient and informed.

Story 2: Data Analyst - Kelly

Kelly starts her day by pulling new satellite images from multiple sources. She

uses GIS software to correct and enhance these images for clarity. By overlaying new images on old ones, she detects significant changes, like deforestation or urban growth. She updates Google Maps with the new data, helping to ensure that users have the latest information when planning routes or exploring areas.

Story 3: Street View Driver - Mario

Mario is responsible for capturing street-level images for Google Maps. His van, equipped with a multi-lens camera, captures 360-degree views of the streets he drives. He spends his day driving assigned routes, often in new cities or areas where significant changes have occurred. The images he captures are sent to Google for processing and integration into the Street View feature.

Story 4: 3D Modeling Specialist - Mei

Mei creates realistic 3D models of buildings and landmarks for Google Maps. She uses high-resolution aerial images and elevation data as her primary sources. Through GIS software, she's able to translate this data into detailed 3D models. When a user explores a city using the 3D feature in Google Maps, they're seeing Mei's work.

Story 5: Software Engineer - Lucas

Lucas works on the backend of Google Maps, making sure everything runs smoothly. His day often involves optimizing algorithms that calculate routes, improving the map rendering speed, or ensuring that the huge amount of data Google Maps uses is stored efficiently. Lucas' work doesn't always have a visible output on the Google Maps interface, but without it, the service wouldn't be nearly as fast or reliable.

COMPONENTS OF INFORMATION SYSTEMS

Every information system despite its complexity is composed of four components: hardware, software, data and people. These four components are used to perform the functions of data collection, storage, processing and sharing.

**AN INFORMATION SYSTEM HAS FOUR PRIMARY COMPONENTS:
HARDWARE, SOFTWARE, DATA, AND PEOPLE.**

Hardware: The physical components of an IS, including computers, network equipment, and IoT devices. Computers include all types personal computers as well as servers, whether they are local servers or part of a data center. Network equipment includes switches and routers. While IoT devices can be a very wide variety of devices that range from simple sensors to smart cars.

The school information system hardware includes all the personal computers of its users as well as the school server and the smart ID carried by students. The geographic information system includes a tremendous number of hardware that range from satellites to every smartphone running the Google Maps app. Also included in the Google Maps information system are the data centers that store and process its data, and share its information with all its users.

Software: The programs and applications used to process and manage information. These also include the databases that contain all the data. Some information systems consist of one application that performs all the functions, whereas others may consist of multiple applications each performing a particular task and collectively forming an information system.

In the case of the school information system, it consists of one application which all users access. Whereas the geographic information system consists

of multiple applications, Google Maps accessible by the general public, as well as data analysis software, photo-editing software and 3D modeling software to name a few.

Data: The raw, unprocessed facts and figures without any added interpretation or analysis. Data is simply the input collected either from users entering it using the application or automatically collected by IoT devices.

School Information Systems (SIS) Data

Student Data: This includes data such as student names, IDs, grades, test scores, attendance, demographic data, and any other information related to individual students.

Class Data: Information about each class, including the teacher, room number, subject, schedule, and enrolled students.

Teacher Data: Details about teachers, such as their name, contact information, subjects taught, schedules, and any other relevant information.

Parent Data: Parent or guardian names, contact information, and their relation to students.

School Calendar: Important dates for the academic year, including holidays, start and end dates of terms, and dates of significant events.

Administrative Data: Information about school operations, including budgets, resource allocation, staff details, and other operational data.

Geographic Information Systems (GIS) Data

Satellite Imagery: Images captured by satellites that provide a top-down view of the Earth's surface.

Aerial Photographs: These are photographs taken from aircraft and drones,

which can often provide more detail than satellite imagery.

Maps: Both digital and paper maps can serve as data for a GIS.

GPS Data: Data from Global Positioning System devices can provide very accurate spatial data.

Street View Data: Images and data collected by vehicles equipped with 360° cameras, like those used for Google's Street View feature.

Environmental Sensor Data: Data from sensors that measure environmental conditions like temperature, humidity, soil moisture, etc., often used in environmental or agricultural applications of GIS.

People: Include both the end users who interact with the information system and the people that perform the tasks that keep the system operational, such as technicians, system administrators and software engineers to name a few.

The school information system includes its users, the students, parents, teachers and principle, as well as system administrators and technicians that help keep the system operational. The geographic information system has a far greater number of users, namely any person who has the app, and many other people that work to keep the system running and updated, such as street car drivers, data analysts, 3D modeling specialists and data engineers.

Information systems play an important role in our personal and professional life and demonstrate the importance of information in the decision-making processes of our lives.

SUMMARY

Information systems perform four primary functions: data collection, storage, processing and sharing of data and information.

The overall goal of information systems is to guide decision-making by collecting data, processing it to extract meaningful information, and to share both data and information with users in such a way as to assist them with making decisions.

Information systems have four primary components: hardware, software, data and people. The hardware includes all IoT devices, personal computers, servers and network hardware such as switches and routers. The software includes the programs and applications that are used to collect, store, process and share data and information. Data is a critical component of an information system and is collected in a database as input from IoT devices and users. It is processed to extract useful information. People are part of the information system. They act as users and sources of data input as well as professionals that ensure the information system operates properly.

There are many types of information systems and each has a purpose. A school information system collects, stores, processes and shares data and information with members of the school community to assist each in making decisions related to education and the educational process. A geographic information system collects, stores, processes and shares data and information with the public to assist people in making decisions related to places and travel.

Career Tips

Software Engineers: These individuals design and build information systems. They are responsible for creating the software components that make up the system.

Information System Managers: They are responsible for overseeing the implementation of new systems, managing IT teams, and allocating resources.

Activity 4.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	An information system is responsible for collecting data.
<u>T</u>	Data processing is used to extract meaningful information.
<u>F</u>	An information system does not share information with its users.
<u>T</u>	An information system may incorporate IoT devices.
<u>F</u>	Cloud storage is not an option for information systems.
<u>T</u>	Any device that collects data can be used as component of an information system.
<u>F</u>	An information system cannot include more than one software.
<u>T</u>	A database must remain secure.
<u>F</u>	People do not play an important role in an information system.
<u>T</u>	Information systems presents information to users to guide their decisions.

Activity 4.2

Answer the questions below.

1. What are the four primary functions of an information system?
 - a) Data collection, data mining, data processing, information sharing
 - b) **Data collection, data storage, data processing, information sharing**
 - c) Data collection, data storage, data analysis, data sharing

2. What is the primary purpose of an information system?
 - a) **To guide decision-making**
 - b) To store data
 - c) To collect data

3. Which of the following is a source of data for a school information system?
 - a) teacher
 - b) smart ID
 - c) **Both a and b**

4. Which of the following is a source of data for a geographic information system?
 - a) satellite
 - b) map
 - c) **Both a and b**

5. What is the primary way information systems help with decision-making?
 - a) By collecting data from users
 - b) **By processing raw data into meaningful information**
 - c) By storing data in databases

Activity 4.3

Answer the questions below.

1. What is the role of a system administrator in a school information system?
 - a) To enter data
 - b) To oversee the system and ensure it runs smoothly
 - c) To analyze data

2. What is the role of a data analyst in a Geographic Information System like Google Maps?
 - a) To drive the Street View vehicle
 - b) To pull new satellite images and enhance them for clarity
 - c) To create 3D models of buildings

3. What are the components of an information system?
 - a) Hardware, software, information
 - b) Software, data, IoT devices, cloud
 - c) Hardware, software, data, people

4. What kind of devices does the 'hardware' component of an information system include?
 - a) servers, personal computers
 - b) IoT devices, routers
 - c) Both a and b

5. What do Geographic Information Systems (GIS) primarily visualize data in the form of?
 - a) Charts
 - b) Maps
 - c) Tables

Activity 4.4

Answer the questions below.

1. What are the main functions of an information system?

The main functions of an information system are to collect data, store data, process data into meaningful information, and share information.

2. What are some types of data stored in a school information system?

A school information system stores various types of data including student data (names, IDs, grades, attendance), class data (teacher, room number, subject, schedule), teacher data, parent data, school calendar, and administrative data (budgets, resource allocation).

3. What are some types of data for a geographic information system?

Satellite images, aerial photographs, maps, GPS data, Street view data, environmental sensor data

Activity 4.5

Answer the questions below.

1. What are the main components of an information system?

The main functions of an information system are to hardware, software,
data, and people.

2. What is the role of IoT devices in information systems?

IoT devices play a crucial role in information systems by automating data
collection. These devices can collect data from their environment and send it
to the system for processing and analysis.

3. What role do people play in an information system?

People are both the end users who interact with the system and the
professionals who keep the system operational. They contribute to data input
and use the output information for various purposes. Professionals such as
system administrators and software engineers ensure smooth operation and
maintenance of the system.

Activity 4.6

Interview an IT Professional: Students can prepare and conduct an interview with a school's IT administrator or a GIS specialist. They can ask about the everyday challenges, maintenance requirements, and the skills needed to operate and manage these systems.

Activity 4.7

Build a Mock School Information System: Students can design and create a basic mock-up of a school information system. They can define what data should be collected (such as student names, grades, attendance), decide how to store it, define methods for processing the data, and consider how it should be shared with users. This activity will help them understand the practical aspects of an information system.

Activity 4.8

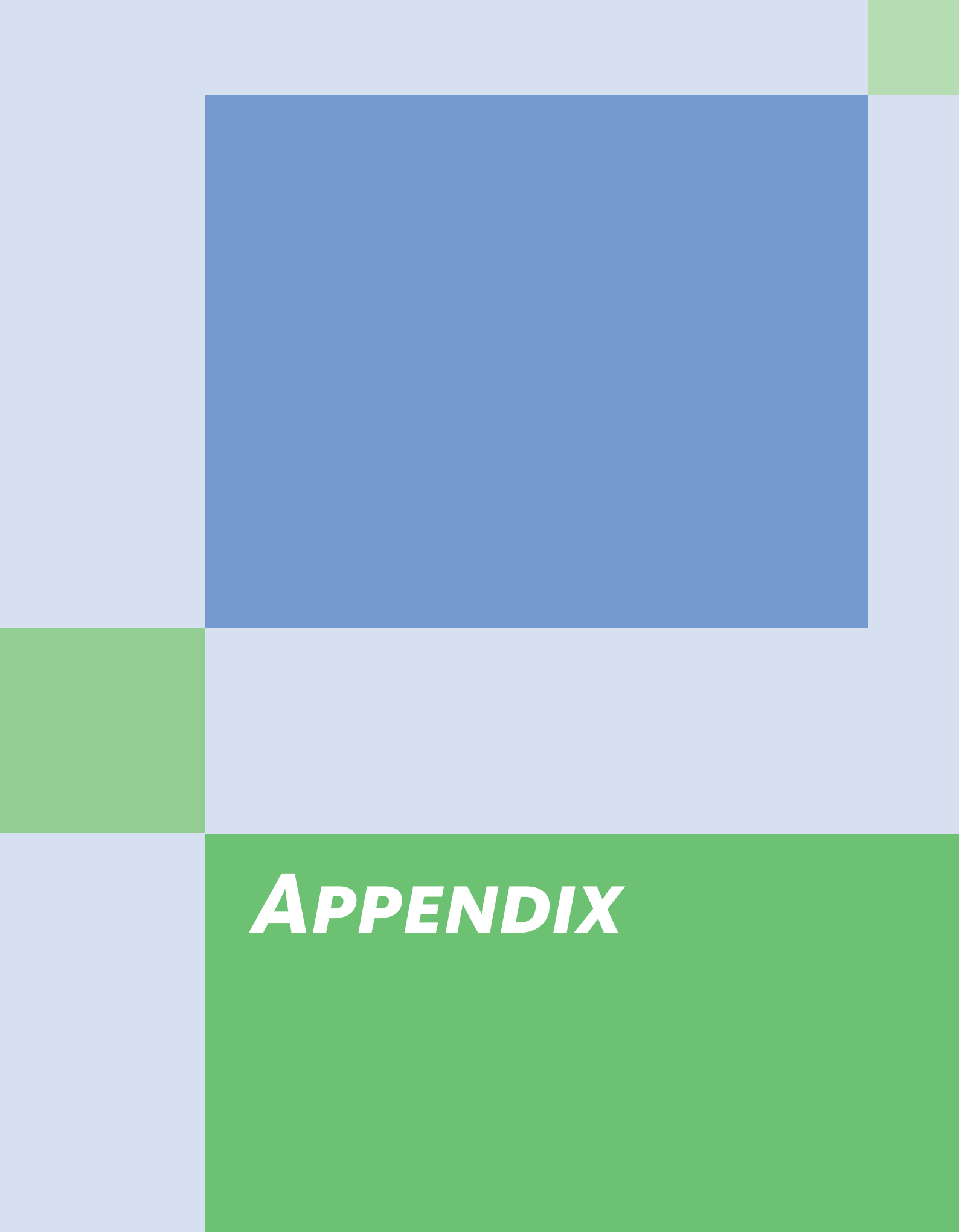
Explore Google Maps: Students can explore the various features of Google Maps (such as Street View, satellite imagery, traffic updates, etc.) and report on how these features help in their daily life. This activity will help them understand how a Geographic Information System works.

Activity 4.9

Discussion on Privacy and Security: Organize a discussion or a debate on the privacy and security concerns related to information systems. It will make students think critically about the ethical aspects of data collection, storage, and sharing.

Activity 4.10

Design User Interface: Students can design the user interface for a new feature in the school information system or Google Maps. They can then present their design to the class, explaining how users would interact with the system via this interface. This activity encourages students to think about the user experience aspect of information systems.

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APPENDIX

GLOSSARY

TERM	DEFINITION	PAGE
Accelerometer	A sensor in devices like smartphones that detects straight-line motion, such as moving up and down, forward and backward, or side to side.	p. 58
Acknowledgement	A signal sent back to confirm that a message was received successfully.	p. 151
Actuator	An output device that performs an action by converting signals into movement, like a motor in a printer or robot.	p. 59
Action Potential	The internal electrical signal used by the human nervous system to transmit information between body parts.	p. 55
Adder	A logic circuit that performs addition on two input signals and outputs their sum.	p. 78
Address	A specific location in memory or registers where data is stored or retrieved.	p. 82
Address Bus	A set of lines in the bus system that carry the addresses of where data should be read from or written to.	p. 103
Administrative Data	Information about how a school or organization runs, such as budgets, staff details, or resources.	p. 198
Aerial Photographs	Pictures taken from aircraft or drones that provide detailed views of the Earth's surface, often used in GIS.	p. 199
Algorithm	A set of steps or rules a computer system follows to process data and solve a problem.	p. 193
Application (App)	A stored program that performs a specific task, such as writing an essay, playing music, or browsing the web.	p. 20–21
Architecture	The basic structure of a computer system, including the processor, memory, input/output devices, a bus, and a clock.	p. 102
Arithmetic Logic Unit (ALU)	The part of the processor that performs calculations and logical operations on data.	p. 81
Autonomous Robot	A robot that can make decisions and act on its own without human control.	p. 128
Bandwidth	The amount of data that can be sent or received in one second across a network, usually measured in bits per second (bps).	p. 175
Binary	A number system using only two values, 0 and 1, which computers use to represent information.	p. 77
Bit	The smallest unit of information in computing, representing either a 0 or a 1.	p. 77

Bluetooth Adapter	A type of adapter that uses short-range radio waves to connect nearby devices, such as headphones, speakers, or keyboards.	p. 146
Bus	A shared communication pathway that connects all components of a computer so they can send and receive signals.	p. 103
Byte	A group of 8 bits that together can represent more complex information.	p. 77
Cache	A small, high-speed memory placed close to the processor to store frequently used data and instructions for quick access.	p. 109
Caesar Cipher	An old encryption method where each letter in a message is shifted by a fixed number of positions in the alphabet.	p. 154
Capability	How powerful a computer is, based on the complexity of tasks it can handle and the speed of its calculations.	p. 34
Checksum	A number calculated from data before transmission, used to check if the data arrived correctly without errors.	p. 152
Class Data	Information about a class, including the teacher, subject, schedule, and enrolled students.	p. 198
Clock	A part of the computer that sends out regular timing signals to keep all components working in sync.	p. 104
Clock Cycle	One complete vibration of the clock's quartz crystal, which acts like a "go" signal for moving data to the next step.	p. 104
Clock Frequency	The number of clock cycles per second; higher frequency usually means faster performance.	p. 106
Cluster	A group of interconnected servers that work together at the same time to handle many user requests in parallel.	p. 175
Command-Line Interface (CLI)	A type of user interface where the user types text commands to make the computer perform tasks.	p. 61
Comparator	A logic circuit that compares two inputs and outputs whether one is greater than, less than, or equal to the other.	p. 80
Communication	The exchange of information between two or more people, often using methods like messaging or email.	p. 18
Computer	A technology designed to process information according to a stored program.	p. 10, 15
Computational Power	The ability of a computer to process large amounts of information and perform calculations very quickly.	p. 34
Concurrent Processing	When a computer quickly switches between tasks to give the impression that multiple tasks are running at the same time.	p. 175
Control Bus	The part of the bus that carries control signals, including timing signals from the clock.	p. 103
Control Unit (CU)	The part of the processor that directs signals, decodes instructions, and activates the correct logic circuits.	p. 84

Core	An individual processing unit inside a CPU or GPU; modern processors may have multiple cores working in parallel.	p. 107
Data	Raw facts or numbers that can be processed to create meaningful information.	p. 15
Data Bus	The part of the bus that carries the actual data being transferred between components.	p. 103
Data Center	A large facility that houses many servers and related systems (cooling, power, security) to provide cloud-based services.	p. 173
Database	An organized collection of data stored on servers that can be accessed and managed by software.	p. 192
Dependability	How well a system, like a data center, can keep running without faults or failures.	p. 177
Destination Address	The location where the result of an operation is stored after processing.	p. 83
Digital Electronic Signal	The electrical signals used inside computers, which can only be high (1) or low (0).	p. 77
Directory (Folder)	A virtual container used by the computer's filing system to organize files; it can also contain other directories (subdirectories).	p. 73
Encryption	The process of changing a readable message into a secret code so that only authorized people can read it.	p. 153
Environmental Sensor Data	Measurements collected by sensors about conditions like temperature, humidity, or soil moisture.	p. 199
Error Detection	Techniques, like checksums and acknowledgements, used to check if data was transmitted correctly.	p. 151–152
Ethernet NIC	A network interface card that connects computers using cables to exchange data.	p. 146
External Bandwidth	The total data transfer capacity between a data center and the outside world (users, internet).	p. 176
Field	A part of a machine-language instruction that carries specific information, such as the opcode or an address.	p. 87
File	A unit of stored information, such as a document, photo, or program, that has a name and address.	p. 19
Filename	The name given to a file so that it can be identified.	p. 19
Frequency-Hopping Spread Spectrum (FHSS)	A method used by Bluetooth to send signals by rapidly switching between radio frequencies to reduce interference.	p. 146
General-Purpose Computer	A computer designed to perform many different tasks using different applications, like desktops, laptops, tablets, and smartphones.	p. 32–33
Geographic Information System (GIS)	A type of information system designed to collect, store, process, and share geographic data, often shown as maps.	p. 195

Google Maps	A widely used GIS application that provides maps, directions, traffic updates, and features like Street View.	p. 194–195
GPU (Graphics Processing Unit)	A processor designed specifically to handle graphics-related instructions, often working alongside the CPU.	p. 107
Graphical User Interface (GUI)	A user interface that uses windows, icons, and menus so people can interact with computers more visually and easily.	p. 62
Gyroscope	A sensor that detects rotation, like rolling, spinning, or tilting movements of a smartphone.	p. 58
Hardware (in IS)	The physical parts of an information system, such as computers, servers, network devices, and IoT equipment.	p. 197
Hop Count	The number of routers a signal must pass through to reach its destination.	p. 147
Human-Computer Interaction (HCI)	A field of study that focuses on how people design, use, and interact with computers and technology.	p. 122
Human-Computer Interface	The boundary where humans and computers exchange signals, usually through input and output devices.	p. 122
Human-Machine Interface (HMI)	The point where humans interact with machines that have embedded computers, like cash registers or smart devices.	p. 126
Information	Any meaningful content that tells us something about ourselves or the world, such as words, images, or sounds.	p. 12
Information System (IS)	A system that collects, stores, processes, and shares data to provide useful information and support decision-making.	p. 192
Input Device (Sensor)	A device that converts signals from the outside world, like light, sound, or pressure, into internal digital signals a computer can use.	p. 57
Input/Output (I/O) Loop	The continuous cycle where a computer's output becomes the human's input, and the human's output becomes the computer's input.	p. 123
Instruction	A command given to the processor that tells it what operation to perform and on which data.	p. 81
Instruction Cache (I-Cache)	A cache memory that stores instructions separately, making it faster for the processor to find the next instruction.	p. 110
Instruction Cycle	The repeated process of fetching, decoding, and executing instructions in a processor.	p. 91
Instruction Set	The complete collection of instructions that a processor is designed to understand and carry out.	p. 83
Interface	The boundary where a system communicates with the outside world by exchanging signals.	p. 56
Internet of Things (IoT)	A system of devices (“things”) connected to a network that can send or receive data automatically without human input.	p. 178
Internal Bandwidth	The data transfer capacity between servers inside a data center.	p. 176

LAN (Local Area Network)	A network covering a small area, like a home, office, or school, where computers and devices share resources.	p. 142
Laptop Computer	A portable personal computer with a built-in screen and keyboard, capable of most tasks a desktop can do.	p. 34–35
Latency	The time it takes for a signal to travel from the source to the destination across a network.	p. 148
Logic Circuit	A collection of transistors arranged to perform a specific operation such as add, multiply, or compare.	p. 78
Machine Language	The language of computers, made up of binary signals (0s and 1s) that represent instructions and data.	p. 85
Magnetometer	A sensor in smartphones that detects the Earth's magnetic field to help determine orientation.	p. 58
Main Memory	The larger but slower memory (RAM) where programs and data are kept for active use.	p. 109
Messaging	A quick way of sending and receiving short pieces of information, often between friends or classmates.	p. 18
Modem	A device that converts digital computer signals into analog signals (and back) so they can travel over telephone lines.	p. 146
Multiplier	A logic circuit that multiplies two input signals and outputs the product.	p. 78
Multiplexer	A logic circuit that selects one signal from several possible inputs and forwards it.	p. 78
Network	A system of interconnected computers and devices that can exchange information.	p. 140
NIC (Network Interface Card)	A hardware card that allows a computer to connect to a network, available in Ethernet, fiber optic, or WiFi versions.	p. 146
Opcode (Operation Code)	The part of an instruction that tells the control unit which operation to perform.	p. 84
Operating System	A group of programs that manage how a computer works and allow other apps to run properly.	p. 20
Output Device	A device that takes internal computer signals and converts them into something external, such as sound, light, or movement.	p. 59
PAN (Personal Area Network)	A small network centered around one person's computer and devices, often using Bluetooth.	p. 140–141
Parallel Processing	When multiple tasks are truly processed at the same time, often by multiple cores or servers.	p. 175
Parallelism	A strategy that speeds up performance by performing multiple tasks at the same time, often using multi-core CPUs or GPUs.	p. 107
Parent Data	Information about students' parents or guardians, such as names and contact details.	p. 198

Performance	A measure of how fast and efficiently a computer completes tasks, influenced by factors like clock speed, cores, and memory design.	p. 106
Peripheral Device	An input or output device connected to a computer to extend its functions, like a printer, headset, or smartboard.	p. 124
Pointer	A small graphical symbol (like an arrow or hand) controlled by a mouse or touchpad to select and move items on the screen.	p. 63
Prediction	A design strategy where the processor guesses what instructions or data will be needed next and fetches them early.	p. 110–111
Prefetching	A prediction method where the processor not only fetches the needed data but also nearby data, assuming it will be used soon.	p. 111
Private Information	Information that is restricted and requires permission or a password to access.	p. 17
Processing (in IS)	The stage where raw data is organized, calculated, or analyzed to turn it into meaningful information.	p. 193
Processor (CPU)	The main component of a computer that changes input into output by executing instructions.	p. 76
Proximity	A design principle where components are placed physically closer together, like cache on the same chip as the processor, to speed up communication.	p. 109
Public Information	Information that is open to everyone and does not require a password.	p. 17
Redundancy	Keeping extra copies of data so it isn't lost if transmission errors occur.	p. 151
Register	A very small and fast memory location inside the processor, used to hold data temporarily.	p. 88
Register File	A group of registers managed together within the processor to store data and addresses close to the ALU.	p. 88, 109
Repeater	A device that boosts weakened signals so they can travel longer distances without losing strength.	p. 151
Response Time	The total time it takes for a user to send a request and receive a response back from a server.	p. 176
Robot	A machine with sensors and actuators that can perform tasks either autonomously or with some level of human control.	p. 128
Router	A device that directs signals between networks, acting as both a gateway to the internet and a traffic controller.	p. 143
Satellite Imagery	Images of Earth taken from satellites, used in GIS to provide wide-area geographic information.	p. 199
School Information System (SIS)	An information system that manages school-related data such as schedules, grades, attendance, and communication with students, parents, and staff.	p. 193
Semi-Autonomous Robot	A robot that can make some decisions on its own but still relies partly on human guidance.	p. 128

Server	A powerful computer in a data center that stores data or runs applications to provide services for other computers and users.	p. 38, 173
Shared Cache	A cache memory used by multiple cores in a processor, storing data that any of them may need.	p. 110
Shielding	A protective layer around wires or circuits to block outside interference.	p. 151
Shifter	A logic circuit that moves the bits of a signal left or right by a certain number of places.	p. 80
Signal	A message carried in physical form, such as electrical energy, that allows parts of a system to communicate and coordinate.	p. 55
Smart Device / Smart Thing	A device with sensors and connectivity that links to a network (like smart watches, fridges, or cameras) to automate or improve tasks.	p. 179
Smart Farming	The use of IoT devices such as soil sensors, drones, and livestock trackers to optimize farming practices.	p. 180
Smart Home	A home equipped with IoT devices (like smart lights, thermostats, or security cameras) connected to a hub that automates daily tasks.	p. 179
Smart Machine	A machine with an embedded computer that adds new features and makes it more functional and user-friendly.	p. 126
Smartphone	A general-purpose personal computer in the form of a phone, capable of running many applications beyond calls and texts.	p. 126
Smart Waste Management	A city system using IoT sensors in bins and trucks to optimize garbage collection and recycling.	p. 181
Software (in IS)	Programs and applications, including databases, that process and manage data within an information system.	p. 197
Source Address	The location where input data for an instruction is stored before processing.	p. 82
Specific-Purpose Computer	A computer designed to perform one set of tasks very well, such as a supercomputer, server, or embedded computer.	p. 36
Specific-Purpose Machine	A machine designed to perform only one type of task, like a cash register at a store.	p. 127
Storage (in IS)	The function of keeping collected data in databases, usually on servers or in the cloud, so it can be accessed later.	p. 192
Storage	The largest and slowest form of memory, used for long-term keeping of data and programs (e.g., hard drives or SSDs).	p. 109
Stored Program	The concept that a computer keeps instructions inside it, allowing it to perform different tasks by running different programs.	p. 19–20
Street View Data	360° street-level images collected by special vehicles for use in GIS platforms like Google Maps.	p. 199
Student Data	Information about students, including names, IDs, grades, test scores, and attendance.	p. 198

Supercomputer	A massive and very powerful computer built to solve one very complex problem at a time, such as weather forecasting or scientific research.	p. 36–37
Synchronization	Keeping all computer components working together in step using timing signals from the clock.	p. 104
System	A set of interconnected parts working together to serve a particular purpose.	p. 54
System Administrator	A professional who oversees and maintains an information system, manages user accounts, installs updates, and ensures smooth operation.	p. 195
Tablet	A portable personal computer that uses a touchscreen instead of a physical keyboard, making it lighter but usually less powerful than a laptop.	p. 34–35
Teacher Data	Information about teachers, such as their names, subjects taught, schedules, and contact details.	p. 198
Technology	Tools, machines, or systems created to help us perform tasks, from pencils to computers.	p. 10
Throughput	A measure of server performance, defined as the number of requests a server can complete in one second.	p. 174–175
Usability	How easy it is for people to use a computer or interface to complete a task.	p. 64
User Interface (UI)	The software part of the human-computer interface that lets users give instructions to a computer and see results.	p. 61
Voice User Interface (VUI)	A user interface where people interact with computers by speaking, using natural language commands.	p. 63
WAN (Wide Area Network)	A larger network covering a city, country, or even the whole globe. The internet is the biggest WAN.	p. 143–144
Web Address	A special type of address used to locate a file or webpage on the World Wide Web.	p. 17
WiFi NIC	A network card that connects a computer to wireless networks using radio signals.	p. 146
WIMP Interface	A type of GUI that uses Windows, Icons, Menus, and a Pointer to interact with the computer.	p. 63
Word (in computing)	The largest group of bits a processor can handle at once; in a 32-bit processor a word is 32 bits, in a 64-bit processor it is 64 bits.	p. 77
World Wide Web (Web)	The collection of files, websites, and content (text, images, videos) that can be accessed through the Internet.	p. 17

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Accelerometer – Sensor in smartphones that detects linear motion in three dimensions: up/down, left/right, forward/backward (p.58).

Actuator – An output device capable of performing an action, such as a motor, valve, or printer head. Converts internal digital signals into external movement (p.59, 72).

Action potentials – Electrical signals used internally by humans to process information (p.122–123).

Adapters – Hardware devices that convert internal digital signals of a computer into external transmission signals (p.145).

- Wired adapters – Land line modems, Ethernet NICs, Fiber optic NICs (p.146).
- Wireless adapters – Cellular modems, WiFi NICs, Bluetooth adapters (p.146).

Adder – A logic circuit that performs binary addition (p.78–79).

Address – Used to locate a file on a computer (p.19). In processors, source and destination addresses specify where data is fetched from or written to (p.82–83, 89–90). See also Web address.

Acknowledgement – Signal sent by a receiver to confirm successful receipt of a transmission (p.152).

ALU (Arithmetic Logic Unit) – The part of the processor responsible for performing arithmetic and logical operations using logic circuits (p.81–82).

Applications (Apps) – Programs stored on a computer that perform specific tasks (p.20–21).

Architecture – The basic structure of a computer, consisting of processor, memory, input/output devices, a bus, and a clock (p.102).

ASCII values – Numerical representation of characters used in checksum calculations (p.166–167).

Attenuation – Weakening of a signal as it travels over long distances (p.150).

Autonomy (robots) – The degree to which a robot can act independently; can be autonomous, semi-

autonomous, or non-autonomous (p.128).

Bandwidth – Amount of data transmitted per second, measured in bits per second (bps); critical for throughput and server performance (p.175–176).

Barcode scanner – Input device used to scan product barcodes, providing information such as name and price (p.127).

Biotechnology – Technology that uses biological processes, cells, and organisms to make products (p.24).

Binary signals – Digital electronic signals that can only have two values: high voltage (1) or low voltage (0) (p.77).

Bluetooth – Wireless short-range communication technology; Class 2 antennas reach 10 m, Class 1 antennas up to 100 m (p.141, 146).

Bus system – A shared communication pathway that connects all components of a computer. Includes the data bus, address bus, and control bus (p.103).

Cache – A memory component placed close to the processor to speed up access to data and instructions (p.109–110).

- Instruction cache (I-cache) – Stores instructions for quick access (p.110).
- Data cache (D-cache) – Stores data items for quick access (p.110).
- Shared cache – Serves multiple cores on a processor (p.110).

Caesar cipher – Early encryption method shifting letters by a fixed number of positions (p.154–155, 168–169).

Capability – The complexity of tasks a computer can perform, determined largely by computational power (p.34).

Cash register – Machine with an embedded computer designed for point-of-sale tasks; includes barcode scanner, keyboard, screen, drawer, and printer (p.126–127).

Checksum – Error detection method where values of transmitted data are summed and compared (p.152, 166–167).

CLI (Command-Line Interface) – A text-based user interface where users type commands to interact with the computer. Common in programming and system administration (p.61).

Clock – Synchronizes activities in a computer by generating regular timing signals. Measured in Hertz (Hz); higher clock frequencies increase performance (p.104–106).

Cloud – System of servers accessed over the internet that provide storage and processing services (p.170–172).

Communication – Exchange of information between two or more people (p.18).

- Email – Formal or longer communication method (p.18, 21).
- Messaging – Quick, short exchanges of information (p.18, 21).

Comparator – A logic circuit that compares two inputs and outputs greater than, less than, or equal (p.80).

Computers – Technology designed to process information according to a stored program (p.10, 15).

- Creating information – Producing new content through writing, recording, or calculation (p.15–16).
- Sharing information – Exchanging content via Internet, messaging, or email (p.16–18).
- Storing information – Keeping files permanently for retrieval (p.19).
- Types of computers – Classified as general-purpose or specific-purpose (p.32).

Control Unit (CU) – Directs the flow of signals through the processor, decodes instructions, generates control signals, and coordinates components (p.84–86).

Cybersecurity – Field of protecting computer systems, networks, and data from unauthorized access and attacks (p.155).

Data – Raw values that can be processed to create new information (example: student grades) (p.15–16).

Data centers – Facilities with large numbers of servers

and supporting systems (switches, cooling, power, security) that handle heavy workloads and provide cloud services (p.38–39, 173–177).

- Dependability – Ability to operate without fault or failure (p.177).
- Performance measures – Throughput, bandwidth, response time (p.174–176).

Data transmission errors – Errors caused by attenuation and interference (p.150).

Database servers – Servers that manage and store data in databases (p.40).

Dependability (systems) – The ability of a system to operate without fault or failure; in data centers can be affected by overheating, outages, breaches, or disasters (p.177).

Desktop computer – Large, stationary personal computer with high capability but low portability (p.34–35).

Digital electronic signals – Electrical signals used by computers for internal information processing. Binary in nature (1 = high voltage, 0 = low voltage) (p.55, 77).

Directory – A logical structure used by the filing system to organize files. Also called a folder; can contain files and subdirectories. Represents a hierarchical structure (p.65, 73–75).

Embedded computers – Computers built into other devices, such as cars, pacemakers, or appliances, designed for energy efficiency, small size, and high reliability (p.41–42).

- Examples – Pacemaker, car engine control, home appliances, airplanes (p.42, 49).
- Other names – Control modules, controllers, microcontrollers (p.49).

Email servers – Handle email accounts and deliver messages (p.40).

Encryption – Securing data by converting plaintext into unreadable ciphertext using an algorithm and a key (p.153–155).

Ethics (robots) – Concerns about whether robots will act responsibly, ethically, and in humans' best interests (p.130).

Execution time – The total time a CPU takes to

complete a program. A measure of performance (p.112, 117).

External drives – Storage devices such as HDDs or SSDs that provide large capacity but risk damage or loss (p.171–172).

Fiber optic cables – Backbone of the internet, transmitting light signals across continents (p.144).

Fields (Instruction) – Sections of a machine language instruction, typically including opcode, data, and addresses (p.87).

File – A unit of storable information (p.19).

- Filename – Label assigned to identify a file (p.19).
- File servers – Servers that store and share files (p.40).

Game servers – Host online games and manage player accounts (p.40).

General-purpose computers – Flexible computers that run a wide range of applications; include personal computers such as desktops, laptops, tablets, and smartphones (p.32–33).

Geographic Information System (GIS) – Information system for collecting, storing, processing, and sharing geographic data, often visualized as maps (p.194–196).

Graphical User Interface (GUI) – Uses windows, icons, and menus to make computers more intuitive and accessible (p.62).

- WIMP Interface – GUI controlled by mouse or touchpad with windows, icons, menus, and pointer (p.63).
- Touchscreen Interface – GUI where users interact directly with the screen through gestures like tapping or swiping (p.63).

GPU (Graphics Processing Unit) – Specialized processor that executes graphics-related instructions. Works in parallel with the CPU (p.107).

Gyroscope – Smartphone sensor that detects rotational movement: clockwise/counterclockwise, rolling, pitching (p.58).

Hardware – The physical components of a computer, including input and output devices, that handle the physical (signal) layer of the interface (p.60).

Hop count – The number of routers a signal passes

through on its path to a destination (p.147–149).

Human-Computer Interaction (HCI) – A multidisciplinary field studying how humans interact with computers and other technologies (p.122).

Human-Computer Interface – The boundary where humans and computers interact via input/output devices (p.122–123).

Human-Machine Interface – The boundary where humans interact with machines, often through embedded computers (p.126).

I-cache (Instruction cache) – Cache memory dedicated to storing instructions for faster execution (p.110).

Information – Meaningful content about ourselves or the world, processed and used by humans and computers (p.12–13).

- Creating (p.15).
- Sharing (p.16–18).
- Storing (p.19).

Information systems – Systems that collect, store, process, and share data to support decision-making (p.192).

- Functions – Collection, storage, processing, sharing (p.192).
- Components – Hardware, software, data, people (p.197–199).
- Examples – School Information Systems (SIS) (p.193–195); Geographic Information Systems (GIS) (p.194–196).

Input devices – Devices that convert external signals (light, sound, pressure) into internal signals; e.g., keyboard, mouse, microphone, camera, touchscreen, sensors (p.57–58).

Input/Output (I/O) loop – Continuous cycle of input and output exchanged between humans and computers until a task is completed or interrupted (p.123).

Instruction cycle – The repeated process of fetch, decode, and execute, performed continuously by the processor (p.91).

Instruction set – The complete set of instructions a processor is designed to perform (p.83, 92).

Interface – The boundary at which a system interacts with the external world by converting signals (p.56).

- User Interface (UI) – Software layer translating user actions into computer instructions (p.61).

Interference – Disruption of signal transmission caused by storms, buildings, electromagnetic radiation, or cosmic rays (p.150).

Integrity of transmission – Assurance that messages are delivered error-free and unchanged (p.149–150).

Internet – A global telecommunications network that connects computers and devices (p.16–17).

Internet of Things (IoT) – System of interconnected devices (“things”) that send or receive data over networks, often autonomously (p.178–181).

- Smart homes – Devices like lights, thermostats, and appliances connected via hubs (p.179).
- Smart farms – IoT used in agriculture with soil sensors, drones, and livestock monitors (p.180).
- Smart cities – IoT for waste management, traffic control, and public safety (p.181).

Keyboard – Input device used to enter text or numbers into a system (p.127).

Laptops – Portable personal computers with capabilities close to desktops, powered by rechargeable batteries (p.34–35).

Latency – The time it takes for a signal to travel from source to destination (p.148–149).

Local Area Network (LAN) – A network covering a limited area like a school or office, usually wired (p.142–143).

Logical layer – The information carried by signals; handled by software in the human-computer interface (p.60).

Logic circuits – Collections of electronic components (mainly transistors) arranged to perform operations such as addition, multiplication, shifting, comparison, and selection (p.78–80).

Magnetometer – Smartphone sensor that detects magnetic fields and orientation relative to Earth’s magnetic field (p.58).

Maps (GIS data) – Digital or paper maps used as part of geographic data (p.199).

Memory cards – Expandable portable storage for devices like smartphones; limited capacity and risk of damage (p.171).

Memory system – Hierarchical system that supplies data and instructions to the processor. Components include storage, main memory, cache, and register file (p.109).

Mobility (robots) – A robot’s ability to move; stationary (factory arms) vs. mobile (vacuums, humanoids) (p.128).

Modems – Devices that convert between digital and analog signals for transmission (p.146).

Multiplier – A logic circuit that multiplies two inputs (p.78–79).

Nanotechnology – Manipulation of matter at the nanoscale (1–100 nm) to make products (p.24).

Networks – Systems of interconnected computers and devices capable of exchanging information (p.140).

- PAN (Personal Area Network) – Smallest network, centered on an individual’s devices (p.140–141).
- LAN (Local Area Network) – Covers homes, offices, or campuses (p.142–143).
- WAN (Wide Area Network) – Largest networks, including the internet (p.143–144).

NIC (Network Interface Card) – Hardware adapter that connects computers to networks (p.146).

Opcode (Operation code) – The part of an instruction that specifies the operation to perform. Decoded by the control unit (p.84–85).

Operating System – Group of stored programs that allow a computer to function properly (p.20).

Output devices – Convert internal signals into external signals (e.g., sound from speakers, light from screens) or actions (via actuators). Includes printers, screens, drawers, projectors, and more (p.59, 125, 127).

Pacemaker – Example of an embedded computer in medicine; regulates heartbeats (p.42).

Parallel processing – Running multiple tasks simultaneously to enhance performance, used in CPUs and clustered servers (p.175–176).

Parallelism – A performance optimization strategy where tasks are performed simultaneously, often

through multi-core CPUs and GPUs (p.107).

Performance measures – Metrics such as clock frequency, instruction count per second, and program execution time (p.106, 112).

Peripherals – External input, output, or I/O devices connected to computers to extend functionality (p.124–125). Examples: printer, projector, smartboard, webcam, scanner, headset.

Personal computers (PCs) – General-purpose computers for individual use, including desktops, laptops, tablets, and smartphones (p.34–35).

- Distinguished by portability (size, weight, power source) and capability (computational power) (p.34).
- Designed to balance flexibility, usability, capability, and affordability (p.35).

Physical layer – The actual signals transmitted and received, handled by hardware (p.60).

Ping – Networking tool to measure round-trip time between source and destination (p.148–149).

Portability – The ease with which a computer can be carried, depending on size, weight, and power supply (p.34).

Prediction – Optimization strategy that anticipates the next instruction or data likely to be needed, using separate caches or prefetching (p.110–111).

Prefetching – A prediction strategy where the processor fetches adjacent data items in advance, improving performance (p.111).

Printers – Output devices producing receipts or documents; use actuators like motors and printheads (p.127).

Processing – Structured change from one form to another, carried out by a series of repeatable steps (p.11–13).

- Information processing – Comparison with memory to identify and interpret (p.13–15).

Processing (cloud-based) – Service where computational power is provided via the cloud rather than on local devices (p.172).

Processor (CPU) – The central unit that changes input into output. Composed of ALU, CU, and register file

(p.76, 92).

Program – A set of instructions to perform a specific task (p.20–21).

- Stored program – Concept that made modern computers possible (p.19–21).

Projector – Peripheral output device that displays visuals for large audiences (p.125).

Public vs. Private Information –

- Public – Accessible to everyone (p.17, 21).
- Private – Restricted by password access (p.17, 21).

Purpose (robots) – The specific tasks robots are designed to perform, from repetitive labor to hazardous work (p.128–129).

Redundancy – Strategy of keeping multiple copies of information to protect against loss (p.151).

Registers / Register file – Small, fast memory components located within the processor, managed by the CU, used to hold data and addresses during execution (p.88–90, 92).

Repeaters – Devices that amplify and retransmit weakened signals to prevent loss (p.151).

Research – Task example for general-purpose computers, such as using a web browser (p.33).

Response time – The time it takes for a server to respond to a user's request, as seen from the user's perspective (p.176).

Robots – Machines capable of performing tasks autonomously or with human intervention, using sensors (input) and actuators (output) (p.128–130).

- Benefits – Perform repetitive or dangerous tasks (p.129).
- Concerns – Job displacement, ethical behavior (p.130).

Routers – Devices that connect LANs to WANs, serving as gateways and traffic controllers; determine best routes using hop count and latency (p.143–149).

School Information Systems (SIS) – Information systems for managing school operations; provide schedules, attendance, grades, communication, and reports (p.193–195).

Screen (cash register) – Output device displaying transaction details (p.127).

Sensors – Input devices that allow robots, IoT, and GIS to collect environmental or situational data (p.128–129, 179–181, 199).

Servers – Computers designed to handle many simpler tasks concurrently, often for sharing or processing information (p.38–40).

- Database servers – Manage and store data (p.40).
- Email servers – Manage accounts and deliver mail (p.40).
- File servers – Store and share files (p.40).
- Game servers – Host games and manage players (p.40).
- Web servers – Host websites and deliver web pages (p.40).
- Characteristics – Optimized for performance, reliability, and security (p.41).

Shielding – Protection for copper wires and satellites against interference (p.151).

Shifter – A logic circuit that shifts bits of input left or right by a specified number of places (p.79–80).

Signals – Messages carried in physical form, such as electrical potential energy in computers (digital electronic signals) or humans (action potentials) (p.55).

Smart machines – Machines with embedded computers that enhance interfaces and add functionality (p.126).

Smartboard – I/O device that displays content and allows direct manipulation (p.125).

Smartphone – A portable general-purpose computer evolved from cellphones. Combines communication (calls, messaging) with computing and internet functions. Considered the smallest, most portable type of personal computer, capable of many tasks (p.34–35, 126).

Software – Programs and databases that determine computer behavior and manage information. In general, software handles the logical layer of the interface (p.60). In information systems, it includes applications and databases that collect, store, process, and share data; these may be single applications or multiple integrated apps (p.197–198).

Specific-purpose computers – Designed for a defined

set of functions, e.g., supercomputers, servers, and embedded computers (p.36–43).

Storage – Permanent repository for information (p.19).

Storage options – Memory cards, external drives (HDD, SSD), and cloud-based storage; each with advantages and limitations (p.171–172).

Street View (GIS data) – Images captured by vehicles with cameras, providing street-level perspectives (p.196, 199).

Supercomputers – Large, powerful computers designed to solve one complex problem at a time (p.36).

- Example – IBM Blue Gene, used for weather forecasting (p.37).
- Uses – Scientific research, weather forecasting, oil and gas exploration, molecular modeling (p.36–37).

Switches – Devices connecting multiple computers and peripherals within a LAN (p.143).

Synchronization – Strategy that ensures components operate in unison using a clock signal, preventing errors and delays (p.104).

Systems – Sets of interconnected components working together for a purpose. Humans and computers are both information-processing systems (p.54).

Tablets – Portable personal computers larger than smartphones but smaller than laptops, using touchscreen technology (p.34–35).

Technology – Tools, machines, or applications of knowledge designed for a purpose (p.10–11).

- Transportation technologies – e.g., car, airplane, ship (p.23–24).
- Other domains – Nanotechnology, Biotechnology (p.24).

Three Laws of Robotics – Ethical rules proposed by Isaac Asimov: robots must not harm humans, must obey humans, and must protect themselves unless it conflicts with the first two laws (p.130).

Throughput – Measure of server performance, defined as the number of requests processed per second (p.174–175).

Transistors – Electronic components that form the

building blocks of logic circuits; their arrangement determines operations (p.78).

Trust in networks – The expectation that transmissions are accurate, secure, and private (p.149–150).

Usability – The ease with which a device or interface can be used to complete a task. Higher in GUIs and touchscreens, lower in CLIs (p.64, 126).

User Interface (UI) – The software component of the human-computer interface, translating input into instructions and output into results (p.61).

- CLI, GUI, VUI – Major types of user interfaces (p.61–63).

Voice User Interface (VUI) – Interface that allows interaction using voice commands, e.g., Siri, Alexa, Google Assistant (p.63).

WAN (Wide Area Network) – Large networks covering cities, countries, or the globe; the internet is the largest example (p.143–144).

Web (World Wide Web) – Internet-based information system of files and pages (p.17).

- Web address – Location identifier for content on the Web (p.17, 19).
- Web servers – Host websites and deliver web pages (p.40).

Webcam / Camera – Input devices that capture images or video, extending the computer interface (p.125).

WiFi – Wireless networking technology for LAN and internet connectivity; typical range 45–150 m (p.146).

WIMP Interface – GUI type using windows, icons, menus, and pointer controlled by mouse/touchpad (p.63).

Word (processor architecture) – The maximum number of bits transmitted simultaneously (e.g., 32-bit word, 64-bit word) (p.77).

